

1. Target Variable

The target variable is **Response**, a binary indicator that equals 1 if the customer accepted the offer in the most recent marketing campaign and 0 otherwise. It is the only variable the model is trained to predict, and the model's output is each customer's estimated probability that Response equals 1.

We chose Response for three reasons. First, it matches the problem statement exactly: the business question is whether a customer will respond to an offer, and Response records that outcome directly. Second, it is a confirmed, observed label for every one of the 2,240 customers, which makes the task a supervised learning problem rather than one that depends on inferred or proxy outcomes. Third, it is the most recent campaign, so it best represents the situation a business faces when deciding whom to contact next.

Response is imbalanced: 334 of 2,240 customers (14.9%) accepted the offer. This shapes the modeling approach. Accuracy would be misleading, because a model that predicts “no response” for everyone would be right about 85% of the time while identifying no one. We will therefore evaluate models with precision, recall, F1-score, and ROC-AUC, and address the imbalance with class weighting or resampling.

The five earlier campaign indicators (AcceptedCmp1 to AcceptedCmp5) are *not* targets. They describe past behavior and are used as predictors. The dataset was originally published for clustering, so it lists no target; we are deliberately repurposing it for classification, and the K-means segmentation in our plan supports interpretation rather than serving as the primary model.

2. Predictors

The predictors are the customer attributes available before an offer is sent, grouped below by the reason each group should help explain who responds.

- **Demographics and household (Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome).** Income and household composition shape both the ability and the need to buy. Age is derived from birth year; total children is derived from Kidhome and Teenhome.
- **Relationship with the company (Dt_Customer, Recency, Complain).** Enrollment date becomes customer tenure. Tenure and days since last purchase capture loyalty and how recently a customer was engaged, while a recent complaint signals dissatisfaction that may reduce responsiveness.

- **Spending behavior (MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds).** Past spending is a direct measure of how much a customer values the company's products, and the mix across categories shows what kinds of offers may appeal. We will also create total spending as a summary measure.
- **Purchasing channel and price sensitivity (NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth, NumDealsPurchases).** Channel use indicates where to reach a customer, and discount purchases indicate sensitivity to offers. We will derive total purchases and preferred channel.
- **Prior campaign engagement (AcceptedCmp1 to AcceptedCmp5).** Past acceptance is likely the strongest single signal of future acceptance. We will also derive a count of prior campaigns accepted.

Excluded variables. ID is only an identifier. Z_CostContact and Z_Revenue are constant across all rows and carry no predictive information. Dt_Customer and Year_Birth are replaced by tenure and age rather than used raw.

Limitations and risks. The spending and purchase columns cover the last two years, and the dataset does not date the last campaign, so some spending may overlap with or follow the response. This creates a possible leakage risk that we will acknowledge as a limitation. In addition, age, marital status, and education are personal attributes that could produce biased targeting; we will assess model behavior across these groups in our fairness analysis.

3. Dataset Exploration

The project uses the Customer Personality Analysis dataset from Kaggle, provided by Dr. Omar Romero-Hernandez. Each row is one customer of a retailer that sells food and wine products, and the columns describe demographics, two years of spending, purchase channels, and responses to marketing campaigns.

3.1 General Statistics

Characteristic	Value
Rows (customers)	2,240
Columns (variables)	29
Numeric variables	26 (25 integer, 1 float)
Text variables	3 (Education, Marital_Status, Dt_Customer)
Missing values	Income only (24 rows)
Constant columns	2 (Z_CostContact, Z_Revenue)

Characteristic	Value
Target class balance	334 responders (14.9%); 1,906 non-responders (85.1%)
Spending window	Last 2 years
Enrollment dates	Approximately July 2012 to June 2014

3.2 Variable Groups

The 29 variables fall into four groups from the dataset's documentation, plus two undocumented constants. Full definitions and data types are in Appendix A.

Group	Variables	Count	Data types
People	ID, Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome, Dt_Customer, Recency, Complain	10	Integer, float, text
Products	MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds	6	Integer
Promotion	NumDealsPurchases, AcceptedCmp1 to AcceptedCmp5, Response	7	Integer
Place	NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth	4	Integer
Other	Z_CostContact, Z_Revenue	2	Integer (constant)

3.3 Data Quality Observations

Initial inspection points to several issues that our data-cleaning weeks will address:

- Income is missing for 24 customers and contains extreme values (a maximum near 666,666), so imputation and outlier handling are needed.
- Year_Birth includes implausible values (as early as the 1890s), which imply ages that are not realistic.
- Marital_Status contains inconsistent or ambiguous categories (for example, "Alone," "Absurd," and "YOLO") that should be consolidated. Education uses labels such as "Graduation" and "2n Cycle" that need standardizing.
- Dt_Customer is stored as text and must be converted to a date before tenure can be calculated.
- The Response class imbalance and the two constant columns, described above, will be handled in modeling and feature selection.

Appendix A. Data Dictionary

Variable	Data type	Definition	Role
ID	Integer	Customer's unique identifier	Excluded
Year_Birth	Integer	Customer's birth year	Predictor (as age)
Education	Text	Customer's education level	Predictor
Marital_Status	Text	Customer's marital status	Predictor
Income	Float	Customer's yearly household income	Predictor
Kidhome	Integer	Number of children in household	Predictor
Teenhome	Integer	Number of teenagers in household	Predictor
Dt_Customer	Text (date)	Date of customer enrollment	Predictor (as tenure)
Recency	Integer	Days since last purchase	Predictor
Complain	Integer (0/1)	Complained in the last 2 years	Predictor
MntWines	Integer	Amount spent on wine, last 2 years	Predictor
MntFruits	Integer	Amount spent on fruit, last 2 years	Predictor
MntMeatProducts	Integer	Amount spent on meat, last 2 years	Predictor
MntFishProducts	Integer	Amount spent on fish, last 2 years	Predictor
MntSweetProducts	Integer	Amount spent on sweets, last 2 years	Predictor
MntGoldProds	Integer	Amount spent on gold products, last 2 years	Predictor
NumDealsPurchases	Integer	Purchases made with a discount	Predictor
AcceptedCmp1	Integer (0/1)	Accepted offer in 1st campaign	Predictor
AcceptedCmp2	Integer (0/1)	Accepted offer in 2nd campaign	Predictor
AcceptedCmp3	Integer (0/1)	Accepted offer in 3rd campaign	Predictor
AcceptedCmp4	Integer (0/1)	Accepted offer in 4th campaign	Predictor
AcceptedCmp5	Integer (0/1)	Accepted offer in 5th campaign	Predictor
NumWebPurchases	Integer	Purchases made through the website	Predictor
NumCatalogPurchases	Integer	Purchases made using a catalogue	Predictor
NumStorePurchases	Integer	Purchases made directly in stores	Predictor
NumWebVisitsMonth	Integer	Website visits in the last month	Predictor
Z_CostContact	Integer	Constant value (3); undocumented	Excluded
Z_Revenue	Integer	Constant value (11); undocumented	Excluded
Response	Integer (0/1)	Accepted offer in the last campaign	TARGET